

FRAA: A Retrieval-Augmented Agent for Explainable Financial Risk Assessment

Abstract. Financial risk assessment requires calibrated predictions, timely external evidence, and explanations that can be inspected by human reviewers. Existing industrial systems often rely on manually engineered tabular features and tree ensembles, which are effective but limited in modeling long-range behavioral sequences and fast-changing market or regulatory context. This paper presents the Financial Risk Assessment Agent (FRAA), a retrieval-augmented agentic framework for explainable financial risk assessment. FRAA encodes heterogeneous user behavior with a time-aware Transformer, generates a query from the current user state, retrieves relevant documents from a financial knowledge base, fuses retrieved evidence through cross-attention, and produces both a calibrated risk score and a concise textual justification. On a proprietary 24-month multi-source dataset with over 20 million users, 1.2 billion interactions, and 15 data sources, FRAA achieves a Log Loss of 0.5163 and Recall@5 of 0.4185, improving Log Loss by 19.6% over a deployed Random Forest baseline and by 9.4% over a non-retrieval Transformer. Ablation studies show that knowledge retrieval is the largest contributor, while retrieval-depth analysis identifies five documents as the best trade-off between evidence coverage and noise. Additional evaluations show strong cross-scenario adaptation, explanation quality rated 4.32/5 by domain experts, and 15.7 ms per-query latency under the tested serving configuration.

Keywords: Financial risk assessment · Retrieval-augmented generation · Explainable AI · Agentic modeling · Time-aware Transformer

1 Introduction

Financial institutions assess user-level risk from heterogeneous and time-varying evidence, including transaction histories, account states, macroeconomic indicators, regulatory disclosures, and news sentiment. Accurate risk estimates support credit review, fraud prevention, margin control, and compliance workflows. In practice, many deployed systems still depend on carefully engineered tabular features and tree-based models such as Random Forests and gradient boosting [1, 2, 8]. These models remain strong baselines, but they do not naturally capture long-range sequential dependencies, and they require repeated manual updates when external conditions change.

Recent neural sequence models and Transformer architectures provide a more flexible way to model user behavior over time [13, 6, 12]. However, a purely parametric sequence model has two limitations for financial risk assessment. First,

it can only use external market or regulatory information that has already been encoded in the training data or engineered features. Second, its predictions are difficult to audit unless supported by faithful explanations. Retrieval-augmented generation (RAG) and dense retrieval methods offer a principled way to condition model decisions on external evidence [7, 9], while explanation and calibration methods show the importance of making model outputs inspectable and reliable [11, 10, 3]. These lines of work motivate a risk model that can retrieve fresh financial evidence and explain its scoring behavior.

We propose the **Financial Risk Assessment Agent (FRAA)**, a retrieval-augmented agentic model for explainable financial risk assessment. FRAA is agentic in a narrow and operational sense: for each scoring request, it constructs a user state, issues a retrieval query to a financial knowledge base, fuses the retrieved evidence with behavioral features, produces a risk probability, and generates a short explanation grounded in the retrieved context and influential feature groups. This definition avoids treating the agent component as a buzzword; the agentic behavior is the structured retrieve-fuse-score-explain decision process used at inference time.

The main contributions are:

- We introduce a retrieval-augmented financial risk model that combines time-aware sequence encoding, external evidence retrieval, cross-attention evidence fusion, calibrated risk scoring, and explanation generation.
- We evaluate FRAA on a large-scale proprietary multi-source dataset using a strict chronological split, showing consistent improvements over deployed tree-based baselines and neural sequence models.
- We provide component ablations, feature-group occlusion, retrieval-depth analysis, scenario adaptation, explanation quality assessment, and latency measurements to test whether the proposed retrieval and explanation components are necessary.

2 Related Work

2.1 Financial Risk Modeling and Tabular Baselines

Credit and financial risk prediction has long used tree ensembles and gradient boosting because they handle heterogeneous tabular features and offer strong empirical performance [1, 2, 8]. In production settings, such models are often supported by feature stores and domain-specific aggregation rules. Their main weakness is not predictive strength in static tables, but the cost of maintaining hand-crafted features and the difficulty of incorporating newly emerging external evidence at inference time.

2.2 Sequential User Modeling

Sequential recommendation and behavior modeling show that Transformer encoders can capture evolving user states more effectively than recurrent models

in many settings [6, 12, 13]. Financial risk assessment has a similar temporal structure: recent transactions, account changes, and market signals form a non-uniform sequence whose relevance depends on timing and context. FRAA follows this direction but adapts it to risk scoring, where calibration and explanation are as important as ranking performance.

2.3 Retrieval-Augmented and Explainable AI

Dense retrieval and RAG methods condition model outputs on non-parametric external evidence [7, 9]. Efficient vector search libraries such as Facebook AI Similarity Search (FAISS) make such retrieval practical at large scale [5]. In regulated domains, model decisions also need to be interpretable and calibrated. Local explanation methods such as LIME and SHAP [11, 10] and calibration analyses [3] motivate our dual-output design: FRAA reports both a risk probability and a natural-language justification grounded in retrieved evidence.

3 Method

3.1 Problem Formulation

For a user u and decision time t , the task is to estimate the probability $\hat{r}_t \in [0, 1]$ that a material adverse event occurs within the next 30 days. The label $y_t \in \{0, 1\}$ denotes whether such an event is observed. The model receives a time-ordered window of behavioral, profile, macroeconomic, and external-document signals available no later than time t . This chronological constraint is used throughout training and testing to avoid future information leakage.

3.2 Multi-Source State Construction

FRAA first maps the available evidence into three feature families. The dynamic transactional context $\mathbf{d}_t \in \mathbb{R}^{D_d}$ summarizes recent transaction amount, frequency, volatility, merchant category, and event timing. The quasi-static profile context $\mathbf{s}_t \in \mathbb{R}^{D_s}$ contains slowly changing user attributes such as account tenure, product holdings, balance ranges, and risk tier. The market and macro context $\mathbf{m}_t \in \mathbb{R}^{D_m}$ contains time-aligned external indicators available before the scoring time.

These vectors are concatenated and projected into a shared hidden space:

$$\mathbf{x}_t = \mathbf{W}_p[\mathbf{d}_t \parallel \mathbf{s}_t \parallel \mathbf{m}_t] + \mathbf{b}_p, \quad (1)$$

where $\parallel$ denotes concatenation, $\mathbf{W}_p$ and $\mathbf{b}_p$ are trainable parameters, and $\mathbf{x}_t \in \mathbb{R}^{D_h}$ is the initial user-state representation.

3.3 Time-Aware Sequence Encoding

Financial behavior is irregularly spaced, so FRAA injects a time-gap encoding into each event representation. Let $\Delta\tau_i$ denote the elapsed time between consecutive events. The input to layer l of the Transformer encoder is updated as

$$\mathbf{h}_i^{(l)} = \text{TransformerBlock}^{(l)} \left(\mathbf{h}_i^{(l-1)} + \mathbf{p}_{\Delta\tau_i} \right), \quad l = 1, \dots, L, \quad (2)$$

where $\mathbf{p}_{\Delta\tau_i}$ is the time-gap embedding. The final state $\mathbf{h}_t^{(L)}$ summarizes the user’s observed behavior up to time t .

3.4 Retrieval-Augmented Evidence Fusion

To incorporate external knowledge, FRAA converts the encoded user state into a retrieval query:

$$\mathbf{q}_t = f_q(\mathbf{h}_t^{(L)}), \quad (3)$$

where $f_q(\cdot)$ is a lightweight multilayer perceptron. The financial knowledge base contains approximately 1.2 million timestamped documents, including regulatory filings, market reports, news articles, and macroeconomic commentaries. Documents are embedded offline and indexed with FAISS. At inference time, FRAA retrieves the top- K documents by cosine similarity:

$$\mathcal{D}_t = \text{topK}_{d \in \mathcal{K}_t} \frac{\mathbf{q}_t^\top \mathbf{e}_d}{\|\mathbf{q}_t\| \|\mathbf{e}_d\|}, \quad (4)$$

where $\mathcal{K}_t$ includes only documents available no later than t , and $\mathbf{e}_d$ is the embedding of document d . This timestamp restriction prevents future documents from leaking into the prediction.

The retrieved document embeddings are fused with the user state using cross-attention:

$$\tilde{\mathbf{h}}_t = \text{CrossAttn} \left(\mathbf{h}_t^{(L)} \mathbf{W}_Q, \mathbf{D}_t \mathbf{W}_K, \mathbf{D}_t \mathbf{W}_V \right), \quad (5)$$

where $\mathbf{D}_t$ stacks the embeddings of retrieved documents. The fused representation $\tilde{\mathbf{h}}_t$ is passed to the scoring and explanation heads.

3.5 Risk Scoring and Explanation Generation

The risk head outputs a calibrated probability:

$$\hat{r}_t = \sigma(\mathbf{w}_r^\top \tilde{\mathbf{h}}_t + b_r). \quad (6)$$

In parallel, the explanation head generates a concise textual justification by attending to the retrieved documents and feature-group importance signals. The training objective combines binary cross-entropy for risk prediction and a supervised explanation-fidelity term:

$$\mathcal{L} = -y_t \log \hat{r}_t - (1 - y_t) \log(1 - \hat{r}_t) + \lambda \mathcal{L}_{\text{exp}}(\hat{e}_t, e_t^{\text{ref}}), \quad (7)$$

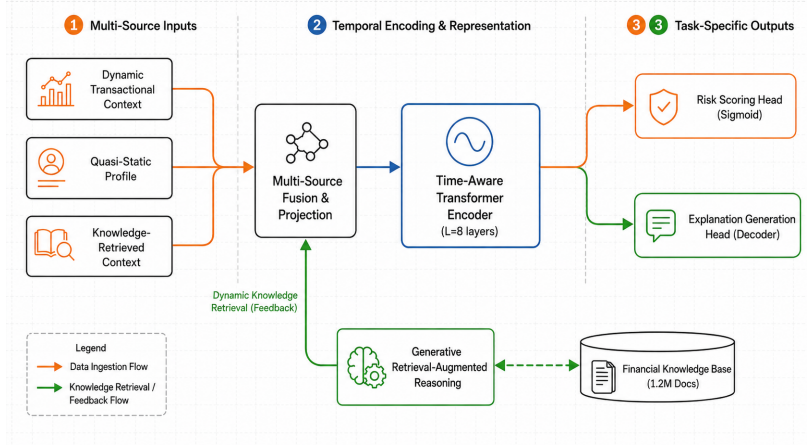


Fig. 1. Overview of FRAA. The model encodes multi-source financial behavior, retrieves timestamp-valid documents from a financial knowledge base, fuses retrieved evidence with the user state, and outputs both a calibrated risk score and an explanation.

where $\hat{e}_t$ is the generated explanation, e_t^{ref} is an expert or rule-supported reference explanation, and λ balances prediction and explanation supervision. In all reported experiments, the Transformer uses 8 attention blocks, hidden dimension 256, and embedding dimension 512; retrieval depth is $K = 5$ unless otherwise stated.

Figure 1 summarizes the full inference path from multi-source evidence encoding to retrieval, fusion, scoring, and explanation generation.

4 Experiments

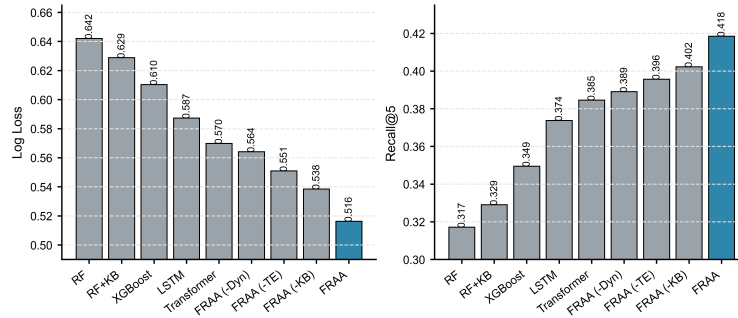
4.1 Dataset and Evaluation Protocol

The dataset is a proprietary 24-month multi-source corpus from a financial risk environment. It contains over 20 million users, 1.2 billion interactions, and 15 heterogeneous data sources. Records include transaction events, account status snapshots, product holdings, macroeconomic series, and timestamped news or regulatory signals. The target is whether a material adverse event, such as default, severe delinquency, or forced liquidation, occurs within the next 30 days.

We use a strict chronological split. Historical records are used to build look-back features and timestamp-valid knowledge-base entries; the model is trained on a 12-month period, tuned on the following month, and evaluated on the final month. No validation or test-period documents are made available to earlier predictions. We report Log Loss for probability calibration, Recall@1/Recall@5 for risk-review ranking, area under the receiver operating characteristic curve (AUC) for feature-group occlusion, and ROUGE-L for explanation overlap with expert-written references.

Table 1. Main test-set performance. Lower Log Loss and higher Recall@5 are better.

Model	Log Loss	Recall@5
RF (deployed baseline)	0.6421	0.3172
RF + KB features	0.6289	0.3291
XGBoost	0.6104	0.3495
LSTM	0.5873	0.3738
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

**Fig. 2.** Main-result comparison. FRAA obtains the lowest Log Loss and highest Recall@5 among all evaluated models.

4.2 Baselines and Training Details

We compare FRAA against a deployed Random Forest (RF), RF with static knowledge-base (KB) features, XGBoost, LSTM, and a non-retrieval Transformer. All baselines use the same chronological split and available feature families. FRAA is implemented in PyTorch and trained with AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-5} , batch size 64, and a maximum budget of 50 epochs. The selected checkpoint is the one with the lowest validation Log Loss.

4.3 Main Results

Table 1 shows that FRAA outperforms both tree-based and neural baselines. Relative to the deployed RF baseline, FRAA reduces Log Loss by 19.6% and improves Recall@5 from 0.3172 to 0.4185. Relative to the non-retrieval Transformer, FRAA reduces Log Loss by 9.4% and improves Recall@5 by 8.8%. The ablated variants remain below the full model, indicating that the retrieval, temporal, and dynamic-feature components contribute complementary information.

Table 2. Cross-scenario adaptation measured by relative Recall lift over scenario-specific baselines.

Adaptation method	Credit card R@1 / R@5	Mortgage R@1 / R@5
Scenario-specific model	0.00 / 0.00	0.00 / 0.00
LP-FT (< 1% params)	+3.14 / +2.97	+2.88 / +2.45
LoRA-FT (< 5% params)	+13.49 / +12.63	+11.72 / +10.88
FRAA full-FT	+18.21 / +17.10	+16.45 / +15.33

Table 3. Component ablation on the risk-score prediction task.

Model variant	Log Loss	Recall@5
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

Figure 2 provides a visual comparison of the same main-result trends reported in Table 1.

4.4 Scenario Adaptation

Table 2 evaluates whether representations learned from the multi-source risk corpus transfer to related financial scenarios. Linear-probe fine-tuning (LP-FT) updates only a small prediction layer, while low-rank adaptation (LoRA) fine-tuning updates low-rank adapter parameters [4]. Full fine-tuning produces the largest gains, while LoRA fine-tuning retains most of the benefit with fewer updated parameters. These results support the claim that FRAA learns reusable cross-product risk representations rather than only fitting a single product-specific task.

5 Ablation and Analysis

5.1 Architectural Components

Table 3 isolates the contribution of dynamic transactions (-Dyn), temporal encoding (-TE), and knowledge retrieval (-KB). Removing the knowledge component causes the largest degradation among FRAA variants: Log Loss increases from 0.5163 to 0.5385 and Recall@5 decreases from 0.4185 to 0.4023. This directly supports the retrieval-augmented design. Removing temporal encoding also hurts performance, confirming the importance of irregular event timing in risk assessment.

Table 4. Feature-group occlusion. A larger AUC drop indicates a more important feature family.

Occluded feature group	AUC drop (%)
Knowledge-retrieved context	3.92
Transaction dynamics	2.87
Macroeconomic indicators	1.55
Quasi-static profile	0.93

Table 5. Effect of retrieval depth K .

Top- K documents	Log Loss	Recall@5
1	0.5287	0.4068
3	0.5214	0.4117
5	0.5163	0.4185
7	0.5175	0.4173
10	0.5189	0.4160

5.2 Feature Groups and Retrieval Depth

The occlusion results in Table 4 are consistent with the component ablation. Knowledge-retrieved context produces the largest AUC drop, followed by transaction dynamics. Static profile variables still contribute, but they are less important than time-sensitive behavioral and external-evidence signals.

Table 5 shows that retrieval depth has a clear optimum. Retrieving only one or three documents provides insufficient external context, while retrieving seven or ten documents slightly degrades both metrics. We therefore use $K = 5$ in the main experiments.

6 Further Evaluation

6.1 Explanation Quality

Five domain experts evaluated 1,000 high-risk test predictions. Table 6 shows that FRAA improves faithfulness, readability, and ROUGE-L over a template baseline that lists the top three feature groups. This supports the value of the explanation head beyond feature enumeration.

6.2 Latency and Offline Business Utility

FRAA is slower than tree ensembles because it performs retrieval and neural sequence encoding, but Table 7 shows that its tested latency remains within an interactive risk-review setting. The retrieval step accounts for approximately 4.2 ms of the total latency.

Table 6. Human and automatic evaluation of generated explanations. Higher scores are better.

Method	Faithfulness	Readability	ROUGE-L
Template-based top-3 groups	3.89	4.02	0.3026
FRAA (full)	4.32	4.51	0.4131

Table 7. Inference latency and throughput under the tested batch-size-64 serving configuration.

Model	Latency (ms/query)	Throughput (queries/s)
RF + KB features	1.9	12,530
XGBoost	2.1	11,200
LSTM	8.3	5,910
Transformer	12.4	4,320
FRAA (full)	15.7	4,070

The offline replay in Table 8 connects calibration improvement with downstream utility under a fixed risk-appetite constraint. We report this as offline decision-support evidence, not as a claim of causal online revenue impact.

7 Limitations and Reproducibility

The evaluation uses a proprietary financial dataset that cannot be publicly released because it contains sensitive user-level risk and transaction information. To support reproducibility within this constraint, we report the data scale, feature families, chronological split, training configuration, retrieval depth, baselines, and ablation settings. The knowledge base is timestamped, and retrieval only uses documents available at or before the scoring time. The business-utility analysis is based on offline replay and should not be interpreted as causal online impact evidence. Future work should add public benchmark evaluations where legally possible and test whether the same retrieval mechanism transfers across institutions.

8 Conclusion

This paper presented FRAA, a retrieval-augmented agentic framework for explainable financial risk assessment. FRAA combines time-aware sequence modeling, timestamp-valid knowledge retrieval, cross-attention evidence fusion, calibrated risk scoring, and explanation generation. On a large-scale proprietary multi-source dataset, FRAA achieves the best Log Loss and Recall@5 among all evaluated models. Ablations identify knowledge retrieval as the most important component, and retrieval-depth analysis shows that five documents provide

Table 8. Offline replay analysis of risk-adjusted revenue (RAR) utility.

Model	Log Loss reduction (%)	RAR lift (%)
RF + KB features	-2.06	+1.89
Transformer	-11.26	+5.34
FRAA (-KB)	-16.13	+7.80
FRAA (full)	-19.59	+10.45

the best trade-off between evidence coverage and noise. Additional evaluations show cross-scenario transfer, improved explanation quality, and practical tested latency. These results indicate that retrieval-augmented evidence can materially improve financial risk assessment when combined with temporal user modeling and auditable explanations.

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